Disciplina: Cálculo Diferencial e Integral I
Conteúdo: Integral

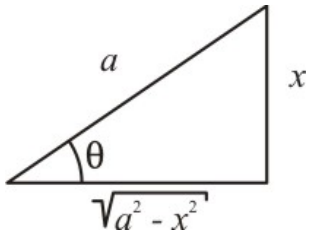
Substituições Trigonométricas

Integrais que envolvem:

a) $\sqrt{a^2 - x^2}$ b) $\sqrt{a^2 + x^2}$ c) $\sqrt{x^2 - a^2}$

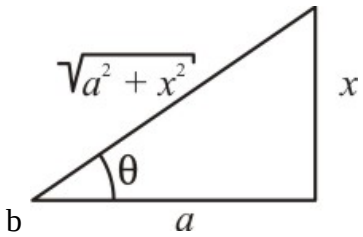
Fazemos:

a)



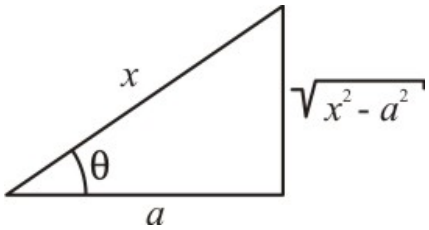
$$\frac{x}{a} = sen\theta \quad \Rightarrow x = asen\theta$$

b)



$$\frac{x}{a} = tg\theta \quad \Rightarrow x = atg\theta$$

c)



$$\frac{a}{x} = \cos\theta \quad \Rightarrow x = a\sec\theta$$

Observação: Depois de integrada, volta-se a variável para x.

Calcule:

1. $\int \frac{\sqrt{4-x^2}}{x}dx$

Solução

$$\int \frac{\sqrt{4-x^2}}{x}\,dx = \int \frac{\sqrt{2^2-x^2}}{x}\,dx$$

Troca: $x = 2\,\text{sen}\,\theta \Rightarrow dx = 2\cos\theta\, d\theta$

Assim:

$$\int \frac{\sqrt{4-x^2}}{x}\,dx = \int \frac{\sqrt{4-4\,\text{sen}^2\theta}}{2\,\text{sen}\,\theta}\cdot 2\cos\theta\, d\theta$$

$$= \int \frac{\sqrt{4(1-\text{sen}^2\theta)}}{\cancel{2}\,\text{sen}\,\theta}\cdot \cancel{2}\cos\theta\, d\theta = \int 2\sqrt{\cos^2\theta}\cdot\cos\theta\, d$$

$$= 2\int \frac{\cos\theta\cdot\cos\theta}{\text{sen}\,\theta}\,d\theta = 2\int \frac{\text{sen}^2+1}{\text{sen}\,\theta}\,d\theta =$$

$$= 2\left[\int \frac{1}{\text{sen}\,\theta}\,d\theta - \int \frac{\text{sen}^2\theta}{\text{sen}\,\theta}\,d\theta\right] =$$

$$= 2\left[\int \text{cossec}\,\theta\, d\theta - \int \text{sen}\,\theta\, d\theta\right] =$$

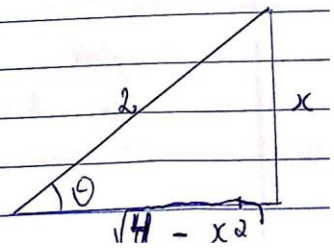
$$= 2\left[-\ln|\text{cossec}\,\theta + \text{cotg}\,\theta| - (-\cos\theta)\right] + K$$

Temos:

- $\text{cossec}\,\theta = \frac{1}{\text{sen}\,\theta} = \frac{1}{(x/2)} = \frac{2}{x}$

- $\text{cotg}\,\theta = \frac{\cos\theta}{\text{sen}\,\theta} = \frac{\left(\frac{\sqrt{4-x^2}}{2}\right)}{\left(\frac{x}{2}\right)} = \frac{\sqrt{4-x^2}}{x}$

- $\cos\theta = \frac{\sqrt{4-x^2}}{x}\,dx$



$$\therefore \int \frac{\sqrt{4-x^2}}{x}\,dx = -2\ln\left|\operatorname{cossec}\theta + \operatorname{cotg}\theta\right| + 2\cos\theta + K =$$

$$= -2\ln\left|\frac{2}{x} + \frac{\sqrt{4-x^2}}{x}\right| + \frac{2\sqrt{4-x^2}}{2} + K =$$

$$= -2\ln\left|\frac{2+\sqrt{4-x^2}}{x}\right| + \sqrt{4-x^2} + K$$

2. $\int \frac{dx}{\sqrt{9+x^2}}$

Resposta: $\ln\left|\frac{\sqrt{9+x^2}+x}{3}\right| + k$

3. $\int \frac{\sqrt{x^2-2}}{x}dx$

Resposta: $\sqrt{x^2-2} - \sqrt{2}arctg\left(\frac{\sqrt{x^2-2}}{\sqrt{2}}\right) + k$

Referências

Guidorizzi, L.H. Um curso de Cálculo – vol 1. 5ª ed. Rio de Janeiro. LTC

Shokowski, E.W. Cálculo com Geometria Analítica. Trad Alfredo Alves de Faria. São Paulo: Makron books, 1994.

$f(x)$	$F(x)=\int f(x)dx$
1	x+k

x^{α}	$\frac{x^{\alpha+1}}{\alpha+1}+k \quad \alpha \neq -1$
$\cos x$	$senx+k$
$senx$	$-\cos x+k$
$\sec^2 x$	$tgx+k$
$\cos\sec^2 x$	$-\cot gx+k$
$\sec xtgx$	$\sec x+k$
$\cos\sec x\cot gx$	$-\cos\sec x+k$
$\frac{1}{1+x^2}$	$arctgx+k$
$\frac{1}{\sqrt{1-x^2}}$	$arcsenx+k$
$-\frac{1}{\sqrt{1-x^2}}$	$\arccos x+k$
$\frac{1}{\sqrt{a^2-x^2}}$	$arcsen\left(\frac{x}{a}\right)+k$
$\frac{1}{a^2+x^2}$	$\frac{1}{a}arctg\left(\frac{x}{a}\right)+k$
$\frac{1}{x}$	$\ln\lvert x\rvert+k$
e^x	e^x+k
$tg(x)$	$-\ln\lvert\cos x\rvert+k$
$\cot g(x)$	$\ln\lvert senx\rvert+k$
$\sec(x)$	$\ln\lvert\sec x+tgx\rvert+k$
$\cos\sec(x)$	$-\ln\lvert\cos\sec x+\cot gx\rvert+k$

Observação: $f(x)=F'(x)$